

STUDENT MANAGEMENT SYSTEM

Project Report

A full-stack web application for managing student records using React, Django REST Framework, and SQLite.

Frontend	React, Axios
Backend	Django, Django REST Framework
Database	SQLite
Version Control	GitHub
Deployment	Render

Prepared as a project / internship report

1. Abstract

The Student Management System is a full-stack web application developed to store and manage student information digitally. The system provides an administrative interface for adding, viewing, updating, searching, filtering, and managing student records. The backend is developed using Django and Django REST Framework, while React and Axios are used for the frontend and API communication. SQLite is used as the database and the application is deployed online using Render.

2. Introduction

Manual management of student records can be time-consuming and may lead to errors. This project provides a centralized digital system for maintaining student details such as name, roll number, email, department, and academic year. The application provides a structured and simple way for an administrator to manage records.

3. Objectives

- Develop a web-based student management application.
- Store student information in a database.
- Implement CRUD operations for student records.
- Provide search and filtering facilities.
- Provide an administrator interface for managing records.
- Deploy the application online for browser access.

4. Technology Stack

Frontend: React and Axios

Backend: Python, Django, Django REST Framework

Database: SQLite

Development: Visual Studio Code

Testing: Postman (recommended)

Version Control: Git and GitHub

Deployment: Render

5. System Architecture

The system follows a client-server architecture. The React frontend sends requests through Axios to the Django REST API. Django processes the request and performs database operations on SQLite. The response is then returned to the frontend.

6. Main Modules

Admin Module – Provides administrator login and access to the Django administration panel.

Student Module – Allows administrators to add, view, update, and delete student records.

Search and Filter Module – Helps locate students by name/details and filter records by department or year.

Database Module – Stores student records using SQLite.

7. Student Information

Each student record contains Name, Roll Number, Email, Department, and Year. In the demonstrated system, a student record was successfully added and displayed in the student list.

8. CRUD Operations

Create – Add a new student record.

Read – View existing student records.

Update – Edit student information.

Delete – Remove a student record.

These operations form the core data-management functionality of the system.

9. Database Design

SQLite is used as the database for the project. A student record contains a unique ID together with name, roll number, email, department, and year fields. SQLite is lightweight and suitable for a small academic project.

10. Deployment

The project is stored in GitHub and deployed as a web service on Render. The deployment dashboard shows the service as Live. Environment variables are configured in Render for production settings and Django administrator configuration.

11. Testing

The application was tested through the Django administration interface. The screenshots demonstrate administrator access, student creation, successful saving of a student, student listing, filtering, and a live deployed instance.

12. Advantages

- Easy student record management
- Reduces manual work
- Fast access to student data
- CRUD functionality
- Search and filtering
- Simple administrative interface
- GitHub-based source management
- Online deployment

13. Limitations

The current version focuses mainly on administrative student record management. Advanced features such as student self-service login, attendance, marks, fee management, notifications, and advanced analytics are not part of the demonstrated version.

14. Future Enhancements

The system can be extended with student login, attendance management, marks and grades, fee management, email notifications, dashboard analytics, PDF/Excel export, role-based access, and a production database such as PostgreSQL for larger deployments.

15. Conclusion

The Student Management System demonstrates the practical use of frontend development, backend development, REST APIs, database management, GitHub version control, and cloud deployment. The implemented Django administration interface provides a functional way to manage student records, while the deployed service makes the project accessible online.

16. Project Links

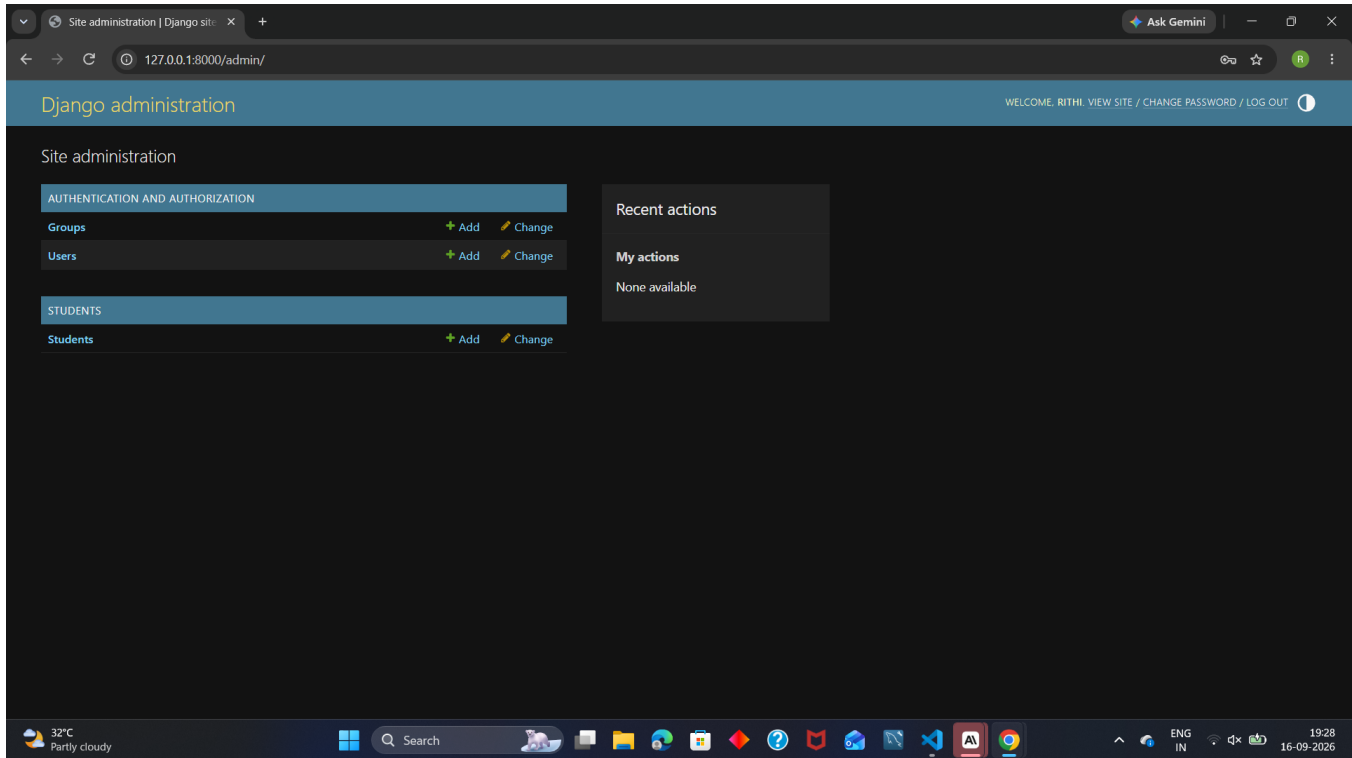
GitHub Repository: github.com/rithikathirupathi-alt/student-management-system

Live Application: student-management-system-muv7.onrender.com/admin/

17. Project Screenshots

The following screenshots document the development, deployment, and working student-management features of the project.

Django Admin Dashboard



GitHub Repository and README

The screenshot displays a web browser window with multiple tabs open, including 'Site admin', 'ChatGP', 'Django', 'GitHub', and 'student'. The active tab shows the GitHub repository page for 'riithikathirupathi-alt / student-management-system'. The URL in the address bar is 'https://github.com/riithikathirupathi-alt/student-management-system/blob/main/README.md'. The repository's navigation bar includes links for 'Code', 'Issues', 'Pull requests', 'Agents', 'Actions', 'Projects', 'Wiki', 'Security and quality', 'Insights', and 'Settings'. On the left sidebar, the 'Files' section is expanded, showing a file tree with folders 'backend', 'frontend', and 'public', and files 'index.html', 'package-lock.json', 'package.json', '.gitignore', and 'README.md'. The main content area displays the 'README.md' file for 'student-management-system'. It includes a 'live link' to 'https://student-management-system-muv7.onrender.com/admin/', a description of the application as a 'full-stack CRUD web application built with React, Django REST Framework, and SQLite', a 'Tech Stack' section listing 'Frontend: React, Axios', 'Backend: Django, Django REST Framework', 'Database: SQLite', and 'API testing: Postman (recommended)', and a 'Project Structure' section. The bottom of the browser window shows a Windows taskbar with a search bar, various application icons, and system status information including '33°C Light rain', 'ENG IN', and the date '17-09-2026'.

student-management-system / README.md

riithikathirupathi-alt Add live link to README cd56deb · 3 minutes ago History

Preview Code Blame 89 lines (73 loc) · 2.6 KB

Student Management System

live link <https://student-management-system-muv7.onrender.com/admin/>

A full-stack CRUD web application built with React, Django REST Framework, and SQLite.

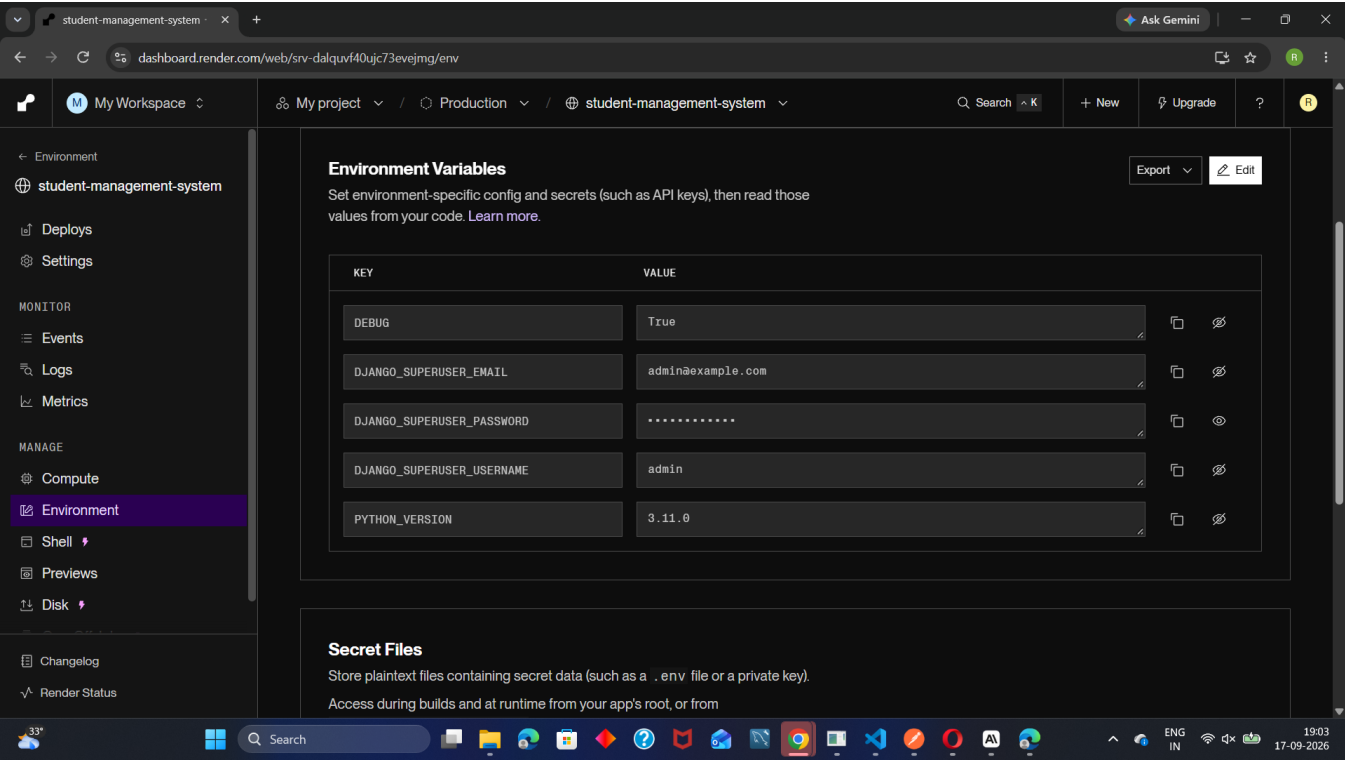
Tech Stack

- Frontend: React, Axios
- Backend: Django, Django REST Framework
- Database: SQLite
- API testing: Postman (recommended)

Project Structure

student-management-system/

Render Environment Variables



Render Deployment Status

Environment

student-management-system

Deploys

Settings

MONITOR

Events

Logs

Metrics

MANAGE

Compute

Environment

Shell

Previews

Disk

Changelog

Render Status

My project / Production / student-management-system

Search

+ New

Upgrade

?

WEB SERVICE

student-management-system

Python 3

Free

Upgrade your instance

Connect

Manual Deploy

Service ID: srv-dalquvf40ujc73evejmg

rrithikathirupathi-alt / student-management-system

main

299cc5c

Live

https://student-management-system-muv7.onrender.com

Live

299cc5c

1m06s

Your free instance will spin down with inactivity, which can delay requests by 50 seconds or more.

Upgrade now

Search deploys and commits

DEPLOY	TRIGGER	DURATION
Update settings.py 299cc5c - Deployed 2min ago by you	Manual	1m06s
Update settings.py 299cc5c - Deployed 2min ago	Build command updated	55.7s Rollback
Update settings.py 299cc5c - Deployed 11min ago	Build command updated	54.9s Rollback

33°

Search

ENG IN

19:07

17-09-2026

Add Student Form

The screenshot displays the Django administration interface for a student management system. The browser's address bar shows the URL `https://student-management-system-muv7.onrender.com/admin/students/student/add/`. The page title is "Django administration" with a user welcome message: "WELCOME, ADMIN. VIEW SITE / CHANGE PASSWORD / LOG OUT".

The left sidebar contains a navigation menu with the following sections:

- Home > Students > Students > Add student
- Start typing to filter...
- AUTHENTICATION AND AUTHORIZATION
 - Groups + Add
 - Users + Add
- STUDENTS
 - Students + Add

The main content area is titled "Add student" and contains the following form fields:

- Name:
- Roll no:
- Email:
- Department: - Year:

At the bottom of the form, there are three buttons: "SAVE", "Save and add another", and "Save and continue editing".

The Windows taskbar at the bottom shows the system clock as 19:32 on 17-09-2026, with a temperature of 31°C and light rain.

Student Added Successfully

The screenshot displays a web browser window with multiple tabs. The active tab is titled "Select student to change | Django". The address bar shows the URL: `https://student-management-system-muv7.onrender.com/admin/students/student/`. The page header is "Django administration" with a user greeting: "WELCOME, ADMIN. VIEW SITE / CHANGE PASSWORD / LOG OUT".

The left sidebar contains a navigation menu with the following sections:

- Home > Students > Students
- Start typing to filter...
- AUTHENTICATION AND AUTHORIZATION
 - Groups + Add
 - Users + Add
- STUDENTS
 - Students + Add

A green success message banner at the top of the main content area reads: "The student 'Janani (33)' was added successfully."

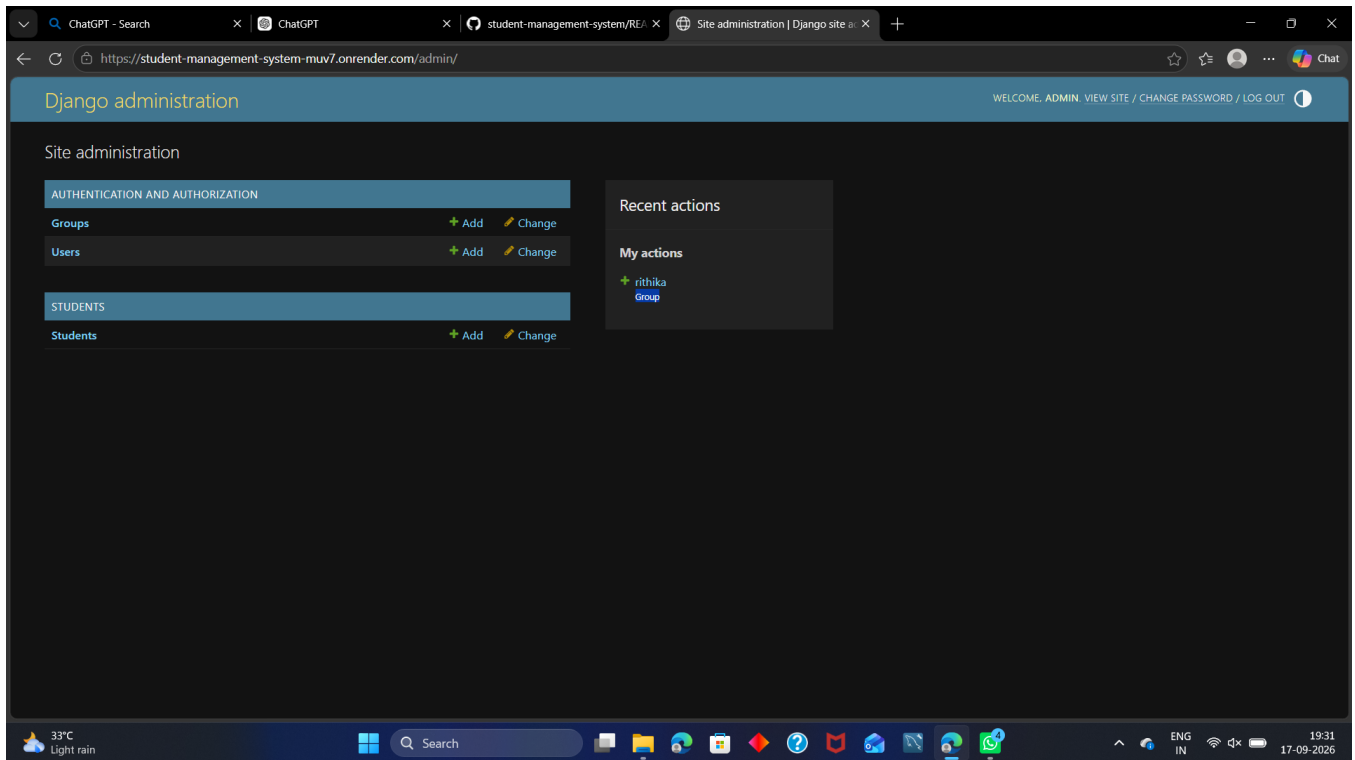
The main content area is titled "Select student to change". It features a search bar and a table of students. The table has columns for ID, NAME, ROLL NO, EMAIL, DEPARTMENT, and YEAR. One student is listed:

ID	NAME	ROLL NO	EMAIL	DEPARTMENT	YEAR
1	Janani	33	jan33@gmail.com	Computer Science	2

Below the table, it indicates "1 student". To the right of the table is a "FILTER" sidebar with options to "Show counts", "By department" (listing All, Computer Science, Electronics & Communication, Mechanical, Civil, and Electrical & Electronics), and "By year" (listing All and 2).

The Windows taskbar at the bottom shows the system clock as 19:32 on 17-09-2026, with a temperature of 31°C and light rain.

Student List and Filtering



Deployed Django Admin

